```
    Tags:
    - Key: Name
      Value: Backend
    UserData:
      'Fn::Base64': |
        #!/bin/bash -ex
        yum -y install httpd                ②
        systemctl start httpd               ③
        echo '<html>...</html>' > /var/www/html/index.xhtml
```

① Launches in the private backend subnet

② Installs Apache from the internet

③ Starts the Apache web server

You're now in serious trouble: installing Apache won't work because your private subnet has no route to the internet. Therefore, the `yum install` command will fail, because the public Yum repository is not reachable without access to the internet.

## *5.5.5 Accessing the internet from private subnets via a NAT gateway*

Public subnets have a route to the internet gateway. You can use a similar mechanism to provide outbound internet connectivity for EC2 instances running in private subnets without having a direct route to the internet: create a NAT gateway in a public subnet, and create a route from your private subnet to the NAT gateway. This way, you can reach the internet from private subnets, but the internet can't reach your private subnets. A NAT gateway is a managed service provided by AWS that handles network address translation. Internet traffic from your private subnet will access the internet from the public IP address of the NAT gateway.